Utku Kalkanli

Computer Vision Engineer
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Available immediately · Full-time

Computer vision engineer: production object detection at Linde, 2D/3D anomaly detection against CAD at Fraunhofer IKS.

Experience

Linde GmbH, Digitalization Oct 2023 – Sep 2026
Machine Learning Engineer Munich, Germany

- Trained and deployed RF-DETR for small-object detection in satellite imagery: mAP@50 0.66 → 0.83. Fully automated an industrial asset-monitoring process; deployed for the sales team.
- Built Airflow pipelines for YOLO and RF-DETR: multi-country COCO dataset preparation, training, ONNX and TensorRT export, Triton-based evaluation, and MLflow experiment tracking.
- Developed CNN and autoencoder models in PyTorch for predictive maintenance, using latent representations for out-of-distribution anomaly detection.
- Maintained and optimized a production Azure ML forecasting pipeline for CPAP therapy adherence (∼1M patients daily). Cut runtime from 37 to 8 hours (78%).
- Contributed to a rotor-orbit fault classifier deployed at a Linde air separation plant, where it predicted a real equipment failure. Named inventor on the invention; co-author of an internal predictive-maintenance whitepaper.

Fraunhofer IKS Oct 2025 – Oct 2026
Research Assistant, Machine Learning (Master's Thesis) Munich, Germany

- Built a CAD-referenced anomaly detector with CLIP/DINOv2 PatchCore feature banks from CAD renders; traced the render-to-real drop to the appearance domain gap.
- Closed part of the gap with depth-conditioned ControlNet SDXL translation and a defect head trained on pixel-exact CAD-defect labels: real-photo AUROC 0.58 → 0.73 without real training images.
- Built a multi-view 3D pipeline: camera pose estimation with Structure from Motion (COLMAP: keypoint detection, feature matching), VGGT where SfM failed on textureless parts, and point-cloud registration to CAD: 94.7% of views registered correctly.

Novit.AI Mar 2022 – Jun 2022
Machine Learning Engineer, Internship Ankara, Türkiye

- Built and deployed a containerized edge AI system (FastAPI, Docker, XGBoost, GitHub Actions, NGINX) with real-time inference and automated rollout to 50+ NVIDIA Jetson devices.

Open Business Software Solutions; Cybersoft 2020 – 2021
Software Engineer, Internships Türkiye

- Built microservices and web apps with Python, Django, PostgreSQL, Elasticsearch, React and TypeScript on GCP.

Education

Technical University of Munich (TUM) Apr 2023 – Oct 2026
M.Sc. in Informatics Munich, Germany
Focus: machine learning, databases, high-performance computing. Thesis: *Reference-Guided Multi-view Anomaly Detection for Industrial Objects* (Prof. Stephan Günnemann, at Fraunhofer IKS).

Bilkent University Sep 2017 – Jun 2022
B.Sc. in Computer Science Ankara, Türkiye

Projects

Radar-Based Ship Identification (TUM industry project with Covadonga Maritime, Nov 2024 – Jun 2025)
Built an automated labeling pipeline matching radar sightings to AIS ship tracks (240 s temporal clustering; one year, 935k sightings, 388 ships), radar-signal features (peak detection, FFT, autocorrelation), and an XGBoost classifier: 91% accuracy on a class-balanced test set. Delivered to Covadonga.

Skills

Computer Vision: Object Detection, Anomaly Detection, Segmentation, 3D Reconstruction, Structure from Motion (SfM), Feature Matching, Keypoint Detection, Pose Estimation, Point-Cloud Registration, Synthetic Data
Models and Libraries: PyTorch, scikit-learn, OpenCV, Open3D, YOLO, RF-DETR, CLIP, DINOv2, VGGT, PatchCore, ControlNet, Stable Diffusion
Deployment: ONNX, TensorRT, NVIDIA Triton Inference Server, CUDA, NVIDIA Jetson, Edge AI, Docker
MLOps and Cloud: MLOps, Apache Airflow, MLflow, Slurm, Ray, Microsoft Azure, Azure ML, Azure DevOps, GitLab
Programming and Hardware: Python, C++, SQL, Bash, Linux, Intel RealSense D435i, Basler cameras
Languages: Turkish (native), English (C1, fluent), German (A2)